

VELOCITY-BASED TRAINING OF LOWER LIMB TO IMPROVE ABSOLUTE AND RELATIVE POWER OUTPUTS IN CONCENTRIC PHASE OF HALF-SQUAT IN SOCCER PLAYERS

JUAN M. RAMÍREZ,¹ VÍCTOR M. NÚÑEZ,¹ CAROLINA LANCHO,^{1,2} MARÍA S. POBLADOR,¹ AND JOSÉ L. LANCHO¹

¹Department of Morphological Sciences, School of Medicine, University of Córdoba, Spain; and ²Rehabilitation Service, University Hospital Donostia, Donostia, Spain

ABSTRACT

Ramírez, JM, Núñez, VM, Lancho, C, Poblador, MS, and Lancho, JL. Velocity-based training of lower limb to improve absolute and relative power outputs in concentric phase of half-squat in soccer players. *J Strength Cond Res* 29(11): 3084–3088, 2015—The power production is force-velocity related. We hypothesized that speed-based training of lower limb using half-squat can lead to absolute and relative power improvements in concentric movement with the same external load. One group of 19 soccer players (age = 24.4 years, $SD = 3.7$ years) participated in a pretest-posttest power training protocol, consistent in 2 training sessions per week during 10 weeks, targeted to work the leg power by performing half-squat with a fixed external load (mean = 71.7, $SD = 5.4$) at 65% of 1 repetition maximum. Measurements of power (absolute [in watt] and relative [in watt per kilogram]), force (in newton), and velocity (in meter per second) (mean and peak) were made from a concentric movement of a half-squat exercise with a fixed external load. The training protocol increased relative power (mean = 47.5, $SD = 47.5$, $p < 0.001$) and absolute power (mean = 169.2, $SD = 95.5$, $p < 0.001$). Also, number of repetitions (mean = 2.9, $SD = 2.4$, $p < 0.01$), force (mean = 66.6, $SD = 36.7$, $p < 0.001$), and velocity (mean = 0.1, $SD = 0.1$, $p < 0.001$) were increased. However, only improved velocity was related to the changes in absolute ($r = 0.939$, $p < 0.001$) and relative ($r = 0.757$, $p < 0.001$) power. The speed-based training combined with moderate to high external load can lead to an improvement of absolute and relative power in concentric phases of half-squat in soccer players. This could be important for improving the performance of the players in the field.

KEY WORDS speed, force, leg, exercise

Address correspondence to José L. Lancho, cm1laalj@uco.es.
29(11)/3084–3088

Journal of Strength and Conditioning Research
© 2015 National Strength and Conditioning Association

3084 ^{the}Journal of Strength and Conditioning Research

INTRODUCTION

Running is the predominant activity of soccer players, yet explosive type actions such as sprints, jumps, duels, and kicking are important factors for successful performance (2). Factors such as acceleration, running, velocity, and capacity to release energy (9) are of major importance (7). Because soccer involves high-intensity intermittent exercise, it has been suggested that a high anaerobic power is a desirable feature of competitive soccer players (1,2). Fitness performance from those who achieved international or professional status shows significant differences for sprint and jump, peak concentric torque, maximal anaerobic power, and other qualities in comparison with those who remained amateur (6). This suggests that improvements in fitness performance, such as power outputs, can play an important role in their chances of promoting higher achievement levels. Moreover, anaerobic power may be as important for performance as for injury prevention (12). The better performance of professional soccer players requires the implementation of specific training methods for athletes, leading to an increase in strength and power parameters, while taking into account the importance of injury prevention. We approach a critical practical issue of this article: it is not necessary to lead the muscle to the limit of its strength to improve the power outputs of the soccer players. Furthermore, this orientation would have a beneficial effect on injury prevention. The selection of exercises that will enhance power and reduce injury is thus a key element of program design for coaches working in soccer.

Power is defined as the rate of performing work, the product of force and velocity, or the rate of transformation of metabolic potential energy to work or heat (17). Maximal muscular power is obtained at optimal values of force and velocity because power is the product of these 2 parameters (19). The ability to generate maximal power during complex motor skills is of paramount importance to successful athletic performance, which is affected by a range of interrelated factors (3), such as heart rate, skin temperature, or blood lactate accumulation

(18). Strength, because an individual cannot possess a high level of power without first being relatively strong (4), and velocity are also related to power performance. Maximal muscular power is determined and limited by the force-velocity relationship and affected by the length-tension relationship.

The effects of speed training methods have been investigated (8), and it seems to be an effective way of improving some segments of power performance in soccer players. However, it is not clear that speed training is more effective than strength training to improve power (20). Different loading schemes can lead to similar improvements in power-related abilities, regardless of the load or velocity used (13). Moreover, the combination of a heavy-resistance and running-speed training program on vertical jump performs better than isolated resistance training without speed training (10). During concentric contraction, muscle shortens, as it contracts. These methods have been shown to be effective in increasing muscle strength, despite eccentric training, which seems to be more successful in promoting such muscle strength gains (15). Concentric half-squat exercise is related with functional outcomes when it is performed against external loading within a range of the load, in the case of which the maximal power output is attained (14). In this sense, the combination of moderate to high load with maximum number of repetitions at a given velocity could be more efficient. Traditionally, the training protocols used for increasing power outputs consisted in a set of exercises, in which athletes lifted heavy weights (nearly 1 repetition maximum [1RM]) with few repetitions. Otherwise, performing a large number of repetitions with low weight was related to the increased endurance. We hypothesized that the combination of both types of training protocols in the same exercise can produce a positive effect on power outputs. This type of power training in which athletes train for a maximal amount of repetitions above a given velocity threshold with a set load would lead to prevent a great number of injuries.

The aim of this study was to determine whether performing maximal amount of repetitions above a predetermined velocity with a predetermined load would lead to increased absolute and relative concentric leg power in soccer players.

METHODS

Experimental Approach to the Problem

To test if velocity-based training, with an external load of 65% of 1RM, increases absolute and relative power outputs, a 1-group pretest-posttest experimental design ($O1 \times O2$) was conducted. Ideally, absolute and relative power increase between pretest and posttest (dependent variables) are due to training protocol. Also, measures of force, velocity, and number of repetitions were taken into account. This approach allowed us to assess the effect of a combined training of strength and endurance on power outputs of the soccer players.

Subjects

Nineteen healthy male soccer players from a Spanish second division professional team volunteered to participate in

a pretest-posttest power training protocol. Before the study, each subject was informed about the nature of the study, reviewing, and signing informed consent forms, which was approved by the Institutional Review Board for the use of subjects of the School of Medicine of the University of Córdoba. One subject had to leave the training program because of injury. Finally, 18 subjects were measured in posttest. The mean age and *SD* (in brackets) of these 18 soccer players was 23.4 (3.7) years (range, 19–31 years), weight was 73.4 (3.6) kg, and height was 179.3 (6.0) cm.

Procedures

Before the implementation of the training program, a structural survey, functional description, and ultrasound examination of the participants were conducted, targeted to rule out any malfunctions. Pretest measures were taken in the Medical Sports Centre of the University of Córdoba on 19 subjects. A 10-week training program protocol with 2 weekly sessions was applied. Posttest measures were taken on 18 survivors. The tests were conducted in the week before and the week after the 10-week training period. All the practice and testing took place at the same time of the day to control any circadian variation in performance. The subjects were asked to participate in 2 training sessions per week (Tuesday and Friday) during 10 consecutive weeks before the soccer training session, targeted to work the leg power by performing half-squats with a fixed external load, which was established as 65% of 1RM. One RM was assessed after the procedure described by Kraemer and Fry (11), consisting of a first warm-up set of 8–10 repetitions at 50% of 1RM. A second warm-up of 3–5 repetitions with ~75% of 1RM was followed, and third warm-up with 90% of 1RM was followed. Finally, the participants performed 1RM exercises by increasing load during consecutive trials. When participants were unable to correctly perform a half-back squat, 1RM was established (mean = 110.38 kg, *SD* = 17.8), which was ranged between 103 and 124 kg (mean = 118.39, *SD* = 19.12). The external load to be used in the training program, which was selected based on 65% of 1RM, was ranged between 65 and 85 kg (mean = 71.7, *SD* = 5.4), representing 97.7% of body weight (Interval of confidence [IC] 95% = 94.0–101.4). Athletes participated in each session, 4 sets of 10–25 repetitions, according to the maximum power output measured in test. When the power level of 2 successive repetitions was below the minimum, the series ended. The feedback was given on sound and movement speed ($\pm 10\%$). The recovery period between each set was 3 minutes. Training sessions lasted 45 minutes and began with a standard 15-minute warm-up: 5 minutes of jogging and 3 sets of 10, 15, and 20 repetitions, respectively of the same test exercise to the 50, 60, and 75% of the corresponding external load. The athletes were instructed not to jump and to keep their heels on the floor.

Measurements

Measurements in pretest and posttest of power (absolute [in watt] and relative [in watt per kilogram]), force (in newton), and velocity (in meter per second) (mean and peak) were

made from a concentric movement of a half-squat exercise with a fixed external load. Participants were encouraged to perform as many repetitions as fast as possible. For 15 seconds, subjects performed successive half-squat repetitions at maximum speed. Before this, they were informed of the correct leg-thigh angle to effect the displacement. Athletes, before measurement, performed a 15-minute warm-up of jogging (5 minutes) and 3 sets of 10, 15, and 20 repetitions, respectively of the same test exercise to the 50, 60, and 75% of the corresponding external load. For measuring power, force, and velocity, a linear encoder connected to Muscle lab-Bosco System (Ergotest Technology-MA.GI.CA.srl, Rome, Italy) was used.

Statistical Analyses

Mean values, *SD*, maximums (Max), and minimums (Min) were used to describe all the variables. To study the efficacy of the training program, 2 paired sample comparisons were computed by means of *t*-test or its nonparametric equivalent, that is, Wilcoxon signed-rank test. Previously, Shapiro-Wilk normality tests were run out for differences between pretest and posttest. Cohen's effect size was calculated by dividing the mean change and *SD* baseline. Pearson's product-moment correlation was also calculated to determine the relationship between selected variables. A level of $p \leq 0.05$ was selected to indicate statistical significance.

RESULTS

Table 1 shows descriptive statistics (Min, Max, mean, *SD*) in pretest and posttest. From pretest to posttest, the number of

repetitions increased to 18.6%, relative power improvement was 8.2%, absolute power increased to 12.6%, force output increased to 4.0%, and velocity increased to 12.5%. The results of the sample paired tests showed a significance change between pretest and posttest in all selected variables (Table 1). The Shapiro-Wilk normality test was applied to these differences (Table 1), showing that Δ Repetitions ($z = 0.834$, $p = 0.005$) and Δ Force ($z = 0.891$, $p = 0.040$) do not follow the normal distribution. In these variables, nonparametric Wilcoxon signed-rank tests were run out, and statistically significant differences were detected between the initial and final measurements (Δ Repetitions: $z = -3.118$, $p = 0.002$, Cohen's $d = 1.93$; Δ Force: $z = -3.636$, $p = 0.000$, Cohen's $d = 0.50$). The Δ Velocity *t*-test indicates an improvement in speed performance between the pretest and the posttest ($t_{17} = 5.426$, $p = 0.000$, Cohen's $d = 1.00$). Regarding the mean power in concentric half-squats, there was also an increase in the relative ($t_{17} = 7.138$, $p = 0.000$, Cohen's $d = 1.00$) and absolute ($t_{17} = 7.303$, $p = 0.000$, Cohen's $d = 0.88$) values between the pretest and the posttest. Cohen's coefficient showed a moderate to large effect size in all variables.

Linear association between Δ variables by Pearson's product-moment correlations (Table 2) shows that there is a positive correlation between the Δ Absolute and Δ Relative power ($r = 0.997$, $p < 0.001$), and between Δ Force and the Δ Repetitions ($r = 0.832$, $p < 0.001$). Of greater interest is the significant correlation between Δ Velocity and Δ Relative power ($r = 0.939$, $p < 0.001$), and between Δ Velocity and Δ Absolute power ($r = 0.939$, $p \leq 0.001$), which indicates that

TABLE 1. Descriptive statistics, differences (posttest minus pretest), Shapiro-Wilk normality tests of differences, and paired samples test with Cohen's effect size.*

	Pretest	Posttest	Dif	SW	St (df)	Cohen's <i>d</i>
Repetitions						
Min–Max	13.0–18.0	14.0–22.0	–4.0 to 6.0			
M (<i>SD</i>)	15.6 (1.5)	18.5 (2.1)	2.9 (2.4)	0.834	–3.118‡	1.93
Relative power ($W \cdot kg^{-1}$)						
Min–Max	14.2–24.0	16.7–26.2	–0.1 to 5.7			
M (<i>SD</i>)	18.2 (2.3)	20.5 (2.6)	2.3 (1.4)	0.967	7.138 (17)§	1.00
Absolute power (W)						
Min–Max	1106.0–1844.4	1208.0–2020.4	–8.0 to 402.6			
M (<i>SD</i>)	1338.6 (192.6)	1507.8 (206.1)	169.2 (95.5)	0.967	7.303 (17)§	0.88
Force (N)						
Min–Max	1444.0–1976.0	1489.4–2013.8	–36.5 to 113.5			
M (<i>SD</i>)	1675.3 (134.5)	1741.9 (127.5)	66.6 (36.7)	0.891	–3.636§	0.50
Velocity ($m \cdot s^{-1}$)						
Min–Max	0.7–1.0	0.7–1.0	0.0 to 0.2			
M (<i>SD</i>)	0.8 (0.1)	0.9 (0.1)	0.1 (0.1)	0.951	5.426 (17)§	1.00

*Dif = difference; SW = Shapiro-Wilk normality test; St = statistic; Max = maximums; Min = minimums; M = mean.

‡ $p < 0.01$.

§ $p < 0.001$.

Variables	1	2	3	4
2. Δ Relative power ($\text{W} \cdot \text{kg}^{-1}$)	-0.128			
3. Δ Absolute power (W)	-0.132	0.997†		
4. Δ Force (N)	0.832†	0.393	0.387	
5. Δ Velocity ($\text{m} \cdot \text{s}^{-1}$)	-0.424	0.939†	0.939†	0.081

2. Cometti, G, Maffiuletti, NA, Pousson, M, Chatard, JC, and Maffulli, N. Isokinetic strength and anaerobic power of elite, subelite and amateur french soccer players. *Int J Sports Med* 22: 45–51, 2001.
3. Cormie, P, McGuigan, MR, and Newton, RU. Developing maximal Neuromuscular power. *Sports Med* 41: 17–38, 2011.
4. Cormie, P, McGuigan, MR, and Newton, RU. Developing maximal Neuromuscular power: Part 2-Training Considerations for improving maximal power production. *Sports Med* 41: 125–146, 2011.
5. Earles, DR, Judge, JO, and Gunnarsson, OT. Velocity training induces power-specific adaptations in highly functioning older adults. *Arch Phys Med Rehabil* 82: 872–878, 2000.
6. Gall, F, Carling, C, Williams, M, and Reilly, T. Anthropometric and fitness characteristics of international, professional and amateur male graduate soccer players from an elite youth academy. *J Sci Med Sport* 13: 90–95, 2010.
7. Hoff, J, Wisloff, U, Engen, LC, Kemi, O, and Hegerud, J. Soccer specific aerobic endurance training. *Br J Sports Med* 36: 218–221, 2002.
8. Jovanovic, M, Sporis, G, Omrcen, D, and Fiorentini, F. Effects of speed, agility, quickness training method on power performance in elite soccer players. *J Strength Cond Res* 25: 1285–1292, 2011.
9. Knuttgen, HG and Komi, PV. Basic Considerations for exercise. In: *Strength and Power in Sport, Volume III of the Encyclopaedia of Sports Medicine* (2nd ed.). P.V. Komi, ed. Hoboken, NJ: Willey-Blackwell, 2003. pp. 3–7.
10. Kotzamanidis, C, Chatzopoulos, D, Michailidis, C, Papaioakou, G, and Patikas, D. The effect of a combined high-intensity strength and speed training program on the running and jumping ability of soccer players. *J Strength Cond Res* 19: 369–375, 2005.
11. Kraemer, WJ, Ratamess, NA, Fry, AC, and French, DN. Strength testing: development and evaluation of methodology. In: *Physiological assessment of humane fitness*, 2nd ed. P. Maud and C. Foster, eds. Champaign, IL: Human Kinetics, 2006. pp. 119–150.
12. Lehance, C, Binet, J, Bury, T, and Croisier, JL. Muscular strength, functional performances and injury risk in professional and junior elite soccer players. *Scand J Med Sci Sports* 19: 243–251, 2009.
13. Loturco, I, Ugrinowitsch, C, Tricoli, V, Pivetti, B, and Roschel, H. Different loading schemes in power training during the pre-season promote similar performance improvements in Brazilian elite soccer players. *J Strength Cond Res* 27: 1791–1797, 2013.
14. Requena, B, González-Badillo, JJ, Saez de Villareal, ES, Erelina, J, García, I, Gapeyeva, H, and Pääsuke, M. Functional performance, maximal strength, and power characteristics in isometric and dynamic actions lower extremities in soccer players. *J Strength Cond Res* 23: 1391–1401, 2009.
15. Roig, M, O'Brien, K, Kirk, G, Murray, R, McKinnon, P, Shadgan, B, and Reid, WD. The effects of eccentric versus concentric resistance training on muscle strength and mass in healthy adults: A systematic review with meta-analysis. *Br J Sports Med* 43: 556–568, 2009.
16. Seger, JY, Arvidsson, B, and Thorstensson, A. Specific effects of eccentric and concentric training on muscle strength and morphology in humans. *Eur J Appl Physiol Occup Physiol* 79: 49–57, 1998.
17. Sub-commission on Publications in the Sports Sciences, IOC Medical Commission. Units of measurement and terminology. In: *Strength and Power in Sport, Volume III of the Encyclopaedia of Sports Medicine* (2nd ed.). P.V. Komi, ed. Hoboken, NJ: Willey-Blackwell, 2003. pp. xiii–xiv.
18. Temfemo, A, Carling, C, and Ahmaidi, S. Relationship between power output, lactate, skin temperature, and muscle activity during brief repeated exercises with increasing intensity. *J Strength Cond Res* 25: 915–921, 2011.
19. Vandewalle, H, Peres, G, Heller, J, Panel, J, and Monod, H. Force-velocity relationship and maximal power on a cycle ergometer: Correlation with the height of a vertical jump. *Eur J Appl Physiol Occup Physiol* 56: 650–656, 1987.
20. Wenzel, RR and Perfetto, EM. The effect of speed versus non-speed training in power development. *J Strength Cond Res* 6: 82–87, 1992.
21. Wisloff, U, Castagna, C, Helgerud, J, Jones, R, and Hoff, J. Strong correlation of maximal squat strength with sprint performance and vertical jump height in elite soccer players. *Br J Sports Med* 38: 285–288, 2004.